14 A teacher asked Tyrese to find the value of $2n^2 - 3n$ when $n = 3$

Here is his working.

$2 \times 3^2 - 3 \times 3$
$= 6^2 - 9$
$= 36 - 9$
$= 27$

(i) What mistake has Tyrese made?

He multiplied 2 and 3 before squaring.

Square 3 first: $2 \times 9 - 9 = 9$.

1 mark
Square before multiplying by 2

(1)

The teacher then asked Megan to find the value of $2n^2 - 3n$ when $n = -4$

Here is her working.

$2 \times -4^2 - 3 \times -4$
$= 2 \times -16 + 12$
$= -32 + 12$
$= -20$

(ii) What mistake has Megan made?

She used -16 for $(-4)^2$, but $(-4)^2 = 16$.

Use brackets: $2 \times 16 - 3 \times (-4) = 44$.

1 mark
Negative squared is positive 16

(1)

(Total for Question 14 is 2 marks)